

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12416465
Kind Code	B1
Date of Patent	September 16, 2025
Inventor(s)	Campbell; Robert Scott

Magazine and adaptor for paintball marker

Abstract

A magazine and a magazine adaptor are described for use with a paintball marker. The paintball marker may include paintball guns and other paintball weapons. The magazine holds paintballs loose in an interior of the magazine for storage, transportation, etc. When the magazine is engaged to the magazine adaptor, a spring-loaded follower places tension on the paintballs. The tension helps to feed the paintballs to the paintball marker in a rapid manner.

Inventors:	Campbell; Robert Scott (St. Louis, MO)
Applicant:	Campbell; Robert Scott (St. Louis, MO)
Family ID:	1000007117021
Appl. No.:	18/198370
Filed:	May 17, 2023

Related U.S. Application Data

us-provisional-application US 63349170 20220606

Publication Classification

Int. Cl.: **F41B11/50** (20130101); **F41B11/52** (20130101); **F41B11/55** (20130101); **F41B11/89** (20130101)

U.S. Cl.:

CPC **F41B11/55** (20130101); **F41B11/50** (20130101); **F41B11/52** (20130101); **F41B11/89** (20130101)

Field of Classification Search

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5809983	12/1997	Stoneking	124/45	F41A 9/82
8966801	12/2014	Bentley	42/50	F41B 11/55
2006/0180134	12/2005	Illuzzi	102/502	F41A 9/24
2011/0226226	12/2010	Isabelle	124/80	F41B 11/52
2011/0271941	12/2010	Meggs	124/31	F41B 11/89
2014/0096755	12/2013	Larmer	124/52	F41B 11/55
2015/0027426	12/2014	Isabelle	124/80	F41B 11/52
2018/0045482	12/2017	Chong	N/A	F41B 11/51
2020/0300575	12/2019	Zuo	N/A	A63H 33/18
2022/0128329	12/2021	Buckmaster	N/A	F41B 11/55

Primary Examiner: Simms, Jr.; John E

Attorney, Agent or Firm: Polsinelli PC

Background/Summary

(1) The present application claims priority U.S. Provisional Patent Application No. (2) 63/349,170, filed Jun. 6, 2022.

FIELD OF INVENTION

(1) The present invention relates to a magazine and adaptor to supply paintballs to a paintball marker or gun.

BACKGROUND OF INVENTION

(2) Paintballs are often held in “pods” which provide a reservoir that holds the paintballs. Paintballs are loaded into “hoppers” that are attached to a paintball marker or gun. The hopper engages to the paintball marker and feeds the paintballs to the marker. Pods and hoppers include an open interior in which the paintballs may move around, spin, rotate, etc. to prevent dimpling, deformation, and/or flat spots on the paintballs. During prolonged storage of paintballs in pods or hoppers, dimpling, deformation, and/or flat spots on the paintballs often occur-particularly when the paintballs rest on top of each other. This is a common problem for paintballs, and a container or package holding the paintballs is often moved or rotated to prevent such deformation.

(3) Paintballs are also held in magazines with spring loading features. Such magazines direct the paintballs to the marker in a rapid manner, however, paintballs stored in such magazines may be subject to the above-described deformation from the tension placed on the paintballs by the spring loading features. And, even if the tension on the paintballs is released when the magazine is not engaged to the marker, the paintballs are still not free to move or rotate to prevent such deformation.

SUMMARY OF INVENTION

(4) A magazine and an adaptor to supply paintballs to a paintball marker or paintball gun is described. The magazine is configured to hold paintballs loose in an interior of the magazine for storage, transportation, etc. The adaptor is configured to engage to the paintball marker or paintball

gun. When the magazine is engaged to the adaptor, tension is placed on the paintballs. The tension helps to feed or direct the paintballs to the paintball marker or gun in a rapid manner. However, when the magazine is not engaged, the paintballs are held loose and horizontally in the magazine in a wide well or chamber and are able to roll freely to avoid dimpling. The magazine serves as a horizontal pod when not engaged to the marker, but the magazine converts to a direct-feed magazine when engaged to the marker. Additionally, when the magazine is engaged, it feeds the paintballs in a wider, “zig-zag” pattern in the magazine chamber to an entry point that funnels the paintballs into the marker, enabling the magazine to rapidly move the paintballs into the marker. The wide well or chamber enables the magazine to allow the paintballs to roll and rotate loosely when the magazine is not engaged, but immediately place tension on the paintballs and assemble them in a double-stack, alternating, “zig-zag” pattern when engaged to the marker. The unique configuration prevents deformation of the paintballs until the magazine is engaged, at which time the paintballs are moved rapidly into the marker.

(5) In one aspect, a magazine and adaptor for a paintball marker is described. The magazine is configured to hold paintballs. An adaptor is configured to engage to a paintball marker. The magazine is configured to engage to the adaptor. The magazine includes a spring-loaded follower. The spring-loaded follower includes a locked configuration and an unlocked configuration. In the locked configuration, the spring-loaded follower is fixed in a position relative to the magazine. In the unlocked configuration, the spring-loaded follower is configured to move relative to the magazine. An engagement of the magazine in the locked configuration to the adaptor transitions the spring-loaded follower to the unlocked configuration.

(6) In another aspect, a magazine and adaptor for a paintball marker is described. The magazine is configured to hold paintballs. An adaptor is configured to engage to a paintball marker. The magazine is configured to engage to the adaptor. The magazine includes a loose configuration and a biased configuration, wherein in the loose configuration, the paintballs are free to move or rotate in an interior of the magazine, and wherein in the biased configuration, the paintballs are urged towards an exit opening of the magazine.

(7) In another aspect, a magazine and adaptor for a paintball marker is described. The magazine is configured to hold paintballs. The adaptor is configured to engage to a paintball marker. The magazine is configured to engage to the adaptor. The magazine includes a spring-loaded follower and a switch configured to release the spring-loaded follower. The adaptor includes a knob extending from an upper surface of the adaptor. The knob is configured to engage the switch when the magazine is engaged to the adaptor.

(8) In another aspect, a magazine for a paintball marker is described. The magazine includes an interior configured to hold paintballs. An opening is at a front of the interior. A spring-loaded follower is positioned at least partially in the interior. The spring-loaded follower is configured to move in the interior toward the opening. A spring-loaded pin is engaged to the magazine. The spring-loaded pin is configured to hold and release the spring-loaded follower at a rear of the interior. A release switch is configured to actuate the spring-loaded pin to release the spring-loaded follower.

Description

BRIEF DESCRIPTION OF DRAWINGS

- (1) FIG. 1A is a perspective view of the magazine and adaptor engaged to the marker.
- (2) FIG. 1B is a sectional view of the magazine and adaptor engaged to the marker.
- (3) FIG. 2 is a perspective view of the magazine engaged to the adaptor.
- (4) FIG. 3 is a lower perspective view of the magazine engaged to the adaptor.
- (5) FIG. 4 is a perspective view of the magazine engaged to the adaptor.

- (6) FIG. 5 is a perspective view of the magazine engaging to the adaptor.
- (7) FIG. 6 is a perspective view of the magazine.
- (8) FIG. 7 is a top view of the magazine.
- (9) FIG. 8 is an exploded view of the magazine.
- (10) FIG. 9 is an exploded view of the magazine.
- (11) FIG. 10 is a lower view of the magazine.
- (12) FIG. 11 is a sectional view of the magazine.
- (13) FIG. 12 is a sectional view of the magazine.
- (14) FIG. 13 is a sectional view of the magazine.
- (15) FIG. 14 is a perspective view of the magazine showing the spring-loaded follower.
- (16) FIG. 15 is a view of the magazine showing the spring-loaded follower in the locked configuration.
- (17) FIG. 16 is a view of the magazine showing the spring-loaded follower in the unlocked configuration.
- (18) FIG. 17 is a view of the adaptor.
- (19) FIG. 18 is an exploded view of the adaptor.
- (20) FIG. 19 is an exploded view of the adaptor.
- (21) FIG. 20 is a sectional view of the adaptor.
- (22) FIG. 21 is a sectional view of the adaptor.
- (23) FIG. 22 is a perspective view of the magazine engaging to the adaptor.
- (24) FIG. 23 is a sectional view of the magazine engaging to the adaptor.
- (25) FIG. 24 is a sectional view of the magazine engaging to the adaptor.
- (26) FIG. 25 is a sectional view of the magazine engaging to the adaptor.
- (27) FIG. 26 is a sectional view of the magazine engaging to the adaptor.
- (28) FIG. 27 is a sectional view of the magazine engaging to the adaptor.
- (29) FIG. 28 is a sectional view of the catch mechanism.
- (30) FIG. 29 is a sectional view of the catch mechanism.
- (31) FIG. 30 is a sectional view of the catch mechanism.

DETAILED DESCRIPTION

(32) For purposes of this application, any terms that describe relative position (e.g., “upper”, “middle”, “lower”, “outer”, “inner”, “above”, “below”, “bottom”, “top”, etc.) refer to an embodiment of the invention as illustrated, but those terms do not limit the orientation in which the embodiments can be used. With respect to FIGS. 1-30, a magazine **100** and a magazine adaptor **400** are described for use with a paintball marker **10**. The paintball marker **10** may include paintball guns and other paintball weapons of all types. The adaptor **400** is configured to removably engage to the paintball marker **10**. The magazine **100** is configured to removably engage to the adaptor **400**.

(33) The magazine **100** is configured to holds paintballs **50** loose in the magazine **100** in a wide well or chamber for storage, transportation, etc. When the magazine **100** is engaged to the magazine adaptor **400**, tension is placed on the paintballs **50**. The tension helps to feed the paintballs **50** to the paintball marker **10** in a rapid manner.

(34) The magazine **100** provides dual holding configurations. The magazine **100** provides a first tensioned or biased configuration and a second loose configuration. In the loose configuration, the magazine **100** acts as a pod in which paintballs **50** are free to move around, spin, rotate, etc. within a wide interior **105** of the magazine **100** to prevent dimpling or other deformation. When the magazine **100** is engaged in the adaptor **400**, a spring-loaded follower **130** may be released to place pressure on the paintballs **50** to feed them into the marker **10** in a rapid manner. The spring-loaded follower **130** immediately assembles the paintballs **50** in a “zig-zag” pattern within the chamber of the magazine **100** and funnels them rapidly into the marker **10**, enabling the magazine **100** to rapidly move the paintballs **50** into the marker **10**. This increases rapid-fire capability for the

paintball marker **10** and loading reliability. Prior to engaging the magazine **100** to the adaptor **400**, the spring-loaded follower **130** may remain locked and place no pressure on the paintballs **50**. This renders the magazine as a “pod” which can store the paintballs **50** without causing dimples or deformation.

(35) With respect to FIG. **7**, the magazine **100** includes a magazine well **108** that loads the paintballs **50** in a double-stack, alternating, “zig-zag” pattern. This increases the number of paintballs **50** that can be loaded and held in a given length of the magazine **100**. It also enables the paintballs **50** to roll and rotate freely in the wider chamber of the magazine **100**, preventing dimpling and deformation. In various aspects, the magazine **100** may be dimensioned to operate with different paintball markers **10** or hold different amounts of the paintballs **50**.

(36) With respect to FIG. **10**, the magazine **100** includes a sliding door **200** to retain the paintballs **50** in the interior **105** of the magazine **100**. The magazine **100** is configured to retract relative to the magazine **100** to cover and uncover the opening **115**. In a closed position, the sliding door **200** covers an opening **115** of the magazine **100**. In an open position, the sliding door **200** does not cover the opening **115**. The sliding door **200** is opened when the magazine **100** is engaged to the adaptor **400**. The sliding door **200** is also opened manually for loading the paintballs **50** into the magazine **100** by depressing the sliding door **200**. However, in this aspect, the sliding door **200** is configured to automatically close when the magazine **100** is removed from the adaptor **400**, thereby keeping the paintballs **50** in the magazine **100**.

(37) The magazine **100** includes the spring-loaded follower **130** that feeds the paintballs **50** to the marker **10** at a rapid pace. The spring-loaded follower **130** travels in the interior **105** of the magazine **100** and is biased toward a front of magazine **100**. In the loose configuration of the magazine **100**, the spring-loaded follower **130** may be locked at a rear of the magazine **100**.

(38) A front portion of the magazine well **108** (at the point of entry into the paintball marker **10**) is horizontally contoured to funnel paintballs **50** into the marker **10** at a rapid pace in an alternating, zig-zag pattern. This enables the magazine **100** to feed a higher quantity of paintballs **50** into the marker **10** than can be fed from a standard, “straight-line” magazine. The horizontally contoured nature of the magazine well **108** provides a feed slope that causes the double-stacked paintballs **50** to become a single stack at the opening **115**.

(39) With respect to FIG. **9**, the magazine **100** includes an upper wall **110** that is generally opposite of middle wall **120**. The upper wall **110** includes a slot **112** generally parallel to a length of the magazine **100**. The slot **112** provides a track for an upper tab **132** of the spring-loaded follower **130**. The user may press against the upper tab **132** in order to retract or move the spring-loaded follower **130** to the locked position. The slot **112** also provides a visual confirmation to the user of the amount of paintballs **50** that are contained in the magazine **100**. The position of the upper tab **132** also provides visual confirmation as whether the paintballs **50** are held in the biased configuration or the loose configuration.

(40) With respect to FIGS. **23-26**, when the magazine **100** is engaged to the adaptor **400**, the sliding door **200** is urged to an open position to allow the paintballs **50** to flow from the magazine **100** into the adaptor **400**. At the same time, the spring-loaded follower **130** unlocks, and the spring-loaded follower **130** biases against the paintballs **50** to drive them forward to the opening **115**. The sliding door **200** is positioned in a recess **220** of the lower wall **140**. The sliding door **200** travels generally parallel to the length of the magazine **100**. The sliding door **200** opens and closes the opening **115** of the magazine **100**.

(41) With respect to FIG. **14**, in this aspect, the spring-loaded follower **130** includes a contact surface **134** that directly contacts the paintballs **50** and pushes the paintballs **50** toward the opening **115**. In this aspect, the spring-loaded follower **130** includes a coil spring **117** to provide bias to the spring-loaded follower **130**. In other aspects, different biasing members may be used to drive or urge the contact surface **134** against the paintballs **50**.

(42) The spring-loaded follower **130** includes the locked and unlocked configurations. In the locked

configuration, the spring-loaded follower **130** is fixed in a position relative to the magazine **100**, and the paintballs **50** are free to rotate, spin, and move. In the unlocked configuration, the spring-loaded follower **130** is configured to move relative to the magazine **100** under the bias of the coil spring **117** and urge the paintballs **50** towards the opening **115** of the magazine **100**.

(43) The adaptor **400** includes an exit opening **500**. In this aspect, the exit opening **500** is located at a lower lateral side of the adaptor **400**. The exit opening **500** is positioned proximate a loading opening **12** of the paintball marker **10** such that paintballs **50** flow from the exit opening **500** of the adaptor **400** to the loading opening **12** of the paintball marker **10**.

(44) With respect to FIGS. **1B** and **17**, a lower surface **402** of the adaptor **400** is shaped and configured to be complementary to various paintball markers **10**. For example, different adaptors **400** may have individualized mating surfaces for specific commercially available paintball markers **10**. For example, different adaptors **400** may locate the exit opening **500** in different positions for specific paintball markers **10**. In this aspect, the lower surface **402** includes a groove **404** configured to lockingly engage an upper portion **14** of the paintball marker **10**.

(45) With respect to FIGS. **23-26**, an upper surface **430** of the adaptor **400** includes a raised entry hole **435**. The raised entry hole **435** includes walls **440** that contact a front surface **205** of the sliding door **200** of the magazine **100**. When the magazine **100** is engaged to the adaptor **400** and the magazine **100** is moved to a forward position, the walls **440** block against the front surface **205** of the sliding door **200** and cause the sliding door **200** to retract, which opens the interior **105** of the magazine **100** and allows flow of the paintballs **50** to the raised entry hole **435**, which leads to a feed tunnel **490** of the adaptor **400**.

(46) In the loose configuration, the magazine **100** acts as a pod in which paintballs **50** are free to move, rotate, spin, etc. In the loose configuration, the spring-loaded follower **130** has been retracted past a spring-loaded pin **300**. In the loose configuration, the spring-loaded pin **300** locks or holds the spring-loaded follower **130** and prevents the spring-loaded follower **130** from moving forward. Thus, the paintballs **50** are free to move around, spin, rotate, etc. within the wide interior **105** of the magazine **100**. With reference to FIG. **14**, the loose configuration of the magazine **100** is illustrated with the spring-loaded follower **130** in the locked configuration. A lower tab **136** of the spring-loaded follower **130** is behind the spring-loaded pin **300**, which holds the spring-loaded follower **130** at the rear of magazine **100**. In the loose configuration, the coil spring **117** is held in a retracted position. The spring-loaded pin **300** is biased inward by a spring **302** and obstructs the spring-loaded follower **130** from traveling forward. The spring-loaded pin **300** travels generally perpendicular to the length of the magazine **100**.

(47) The upper surface **430** of the adaptor **400** further includes a knob **450**, which releases the spring-loaded follower **130**. The knob **450** extends upward from the upper surface **430** of the adaptor **400**. The knob **450** may include a protuberance, extending member, or other structure that projects upward from the upper surface **430** of the adaptor **400**. When the magazine **100** is engaged to the adaptor **400**, the knob **450** fits into a channel **142** of the lower wall **140** of the magazine **100**. As the magazine **100** is moved to the forward position, the channel **142** moves past the knob **450** and the knob **450** eventually engages a release switch **144** for the spring-loaded follower **130**. When the release switch **144** is engaged, the spring-loaded follower **130** is unlocked and drives forward putting bias or tension on the paintballs **50** in the magazine **100**. The magazine **100** is now in the tensioned or biased configuration.

(48) The release switch **144** is movably held in or proximate to the channel **142** of the lower wall **140** of the magazine **100**. The release switch **144** is biased by a spring **145**. The release switch **144** travels parallel to a length of the magazine **100**. In this aspect, the release switch **144** is biased forward towards the front of the magazine **100** by the spring **145**.

(49) When the magazine **100** is moved forward on the adaptor **400**, the knob **450** contacts a front surface **146** of the release switch **144**, thus driving the release switch **144** towards the rear of the magazine **100**. A rear surface **148** of the release switch **144** eventually contacts and drives against

the spring-loaded pin **300**, which causes the spring-loaded pin **300** to retract against the bias from the spring **302**. When the spring-loaded pin **300** sufficiently retracts to clear the spring-loaded follower **130**, the spring-loaded pin **300** may drive forward under the bias of the coil spring **117**. (50) In this aspect, the rear surface **148** of the release switch **144** includes an angled surface which drives against an angled surface of a lower portion **310** of the spring-loaded pin **300**. The rear surface **148** of the release switch **144** and the lower portion **310** of the spring-loaded pin **300** may include complementary camming surfaces.

(51) Thus, in the first or tensioned/biased configuration, the spring-loaded follower **130** is unlocked and is free to move under the bias of the coil spring **117** and urge against the paintballs **50**. In the second or loose configuration, the spring-loaded follower **130** is locked by the spring-loaded pin **300** and does not urge against the paintballs **50**. As described above, the engagement of the magazine **100** (with spring-loaded follower **130** in the locked position) to the adaptor **400** may automatically unlock the spring-loaded follower **130**.

(52) When the magazine **100** is disengaged from the adaptor **400**, the sliding door **200** may automatically close. As the magazine **100** is move from the forward position to a rear position, the walls **440** of the raised entry hole **435** of the adaptor **400** no longer block against the front surface **205** of the sliding door **200**, which permits the sliding door **200** to slide towards the closed position and eventually close. Thus, the paintballs **50** do not pour out of the magazine **100** when the magazine **100** is disengaged from the adaptor **400**.

(53) The magazine **100** includes lateral walls **160** and **170**, which partially define the interior **105** and the magazine well **108** of the magazine **100**. The lateral walls **160** and **170** extend upward from the middle wall **120** of the magazine **100**. The lateral walls **160** and **170** taper inwards towards a front of the magazine **100**. As seen in FIG. 7, the lateral walls **160** and **170** include angled portions **162** and **172** that taper inward toward the opening **115**. This reduces a width of the interior **105** and a width of the magazine well **108** toward the front of the magazine **100**. This inward tapering causes the paintballs **50** to merge from a double-stack arrangement to a single-stack arrangement as the paintballs **50** approach the opening **115**. This arrangement reduces delay or lag in the feeding of the paintballs **50** to the paintball marker **10**. In the double-stack pattern, adjacent paintballs **50** are generally positioned off-set from each other and are not side-by-side.

(54) In this aspect, the magazine **100** is configured to be used with typical paintballs **50**, which have a diameter of approximately 0.68 inches. In this aspect, a majority of the magazine well **108** has a width of approximately 1.0 to approximately 1.2 inches, which is greater than the diameter of the typical paintball. If the width is larger than approximately 1.4 inches, then the paintballs **50** may lodge or jam in the magazine well **108**—as the paintballs **50** may stack in a side-by-side fashion. As such, a majority of the magazine well **108** has a width larger than the diameter of the paintball **50** but smaller than double the diameter of the paintball **50**. Of course, one of ordinary skill in the art will recognize that the dimensions of the magazine **100** may be varied or scaled depending upon the size of the paintballs **50** or specific paintball marker **10**.

(55) As shown in FIG. 7, when the magazine **100** is engaged to the paintball marker **10**, the magazine **100** holds the paintballs **50** in a double-stack, alternating (or “zig zag”) pattern. The spring-loaded follower **130** urges against the paintballs **50** at the rear of the magazine well **108**, which, in turn, causes each paintball **50** to press against the adjacent paintballs **50**. For example, with respect to FIG. 7, each paintball **50** presses against a first paintball **50** to its front and a second paint ball to its front lateral side (at an approximate 45 degree angle). This urges the double-stack of paintballs **50** towards the opening **115**. The angled portions **162** and **172** taper inward toward the opening **115**, which causes the paintballs **50** to merge from the double-stack arrangement into a single-stack arrangement as the paintballs **50** approach the opening **115**. The force from the spring-loaded follower **130** is transferred to and from the paintballs **50** to urge the paintballs **50** from the magazine **100** and through the adaptor **400**. The force from the spring-loaded follower **130** is transferred to and from the paintballs **50** urges the paintballs **50** through the opening **115**, into the

raised entry hole **435** of the adaptor **400**, down the feed tunnel **490** of the adaptor **400**, through the exit opening **500** of the adaptor **400**, and to the loading opening **12** of the paintball marker **10**. The force or pressure from the spring-loaded follower **130** is transmitted through the adjacent paintballs **50** to feed the paintball marker **10** with paintballs **50** in a rapid and consistent manner.

(56) When the spring-loaded follower **130** is in the loose configuration, the spring-loaded follower **130** is locked in place at a rear of the magazine **100**. When the spring-loaded follower **130** is unlocked and is in the tensioned/biased configuration, the spring-loaded follower **130** moves toward the front of the magazine **100**. The spring-loaded follower **130** generally travels between the lateral walls **160** and **170**. The upper wall **110** generally joins the lateral walls **160** and **170** and closes the interior **105** of the magazine **100**. In this aspect, the spring-loaded follower **130** generally travels parallel to a barrel or firing direction of the paintball marker **10**.

(57) The magazine **100** includes lateral extending members **250**, **252**, **254**, **256**. The lateral extending members **250**, **252**, **254**, **256** extend outward from the magazine **100**, i.e., generally perpendicular to the length of the magazine **100**. The lateral extending members **250**, **252**, **254**, **256** assist in aligning the magazine **100** to the adaptor **400**. When the magazine **100** is engaged to the adaptor **400**, the lateral extending members **254** and **256** slide under forward flanges **410** and **412** of the adaptor **400** and lateral extending members **250** and **252** slide under rear flanges **414** and **413** of the adaptor **400**. This engagement between the lateral extending members **250**, **252**, **254**, **256** and the flanges **410**, **412**, **413**, and **414** hold the magazine **100** to the adaptor **400** and prevent an upward separating movement of the magazine **100** from the adaptor **400**. Lateral walls **420** and **422** of the adaptor **400** are sized to contain the lateral extending members **250**, **252**, **254**, and **256**.

(58) With respect to FIGS. **28** and **29**, the upper surface **430** of the adaptor **400** further includes a catch mechanism **470**, which prevents the disengagement of the magazine **100** from the adaptor **400**. When the magazine **100** is moved to the forward position, a contact surface **472** of the catch mechanism **470** lodges in a catch opening **126** in the lower wall **140** of the magazine **100**. The catch mechanism **470** is spring-loaded via spring **476**, such that the contact surface **472** is urged upward by the spring **476**. This engagement between the catch mechanism **470** and the catch opening **126** prevents a lateral separating movement of the magazine **100** from the adaptor **400**. In this aspect, either of levers **480** or **482** of the catch mechanism **470** are actuated to release the magazine **100** from the adaptor **400**. When either of the levers **480** or **482** are actuated, the contact surface **472** is retracted downward to release from the catch opening **126**. Now, the magazine **100** may be slid to the rear position and disengage the lateral extending members **250**, **252**, **254**, and **256** from the forward flanges **410** and the rear flanges **413**.

(59) In this aspect, the user of either levers **480** or **482** provide ambidextrous operation. In this aspect, outer portions **481** and **483** of the levers **480** and **482** are pulled or pushed forward, which rotates inner portions **484** and **485** of the levers **480** and **482** backward and against a lower portion **474** of the catch mechanism **470**. The contact of the inner portions **484** or **485** against the lower portion **474** overcomes the upward bias of the spring **476** and drives the contact surface **472** of the catch mechanism **470** downward. The levers **480** and **482** rotate about axis **486** and **487**, respectively.

(60) The front of the magazine well **108** (at the point of entry into the marker **10**) is also vertically contoured to force-feed paintballs into the opening **115** by the combination of the 45 degree angled pressure of the next paintball **50**, and the contour of the feed slope of the magazine well **108**, working together. The vertically contoured feed slope approaching the opening **115** causes the double-stacked paintballs **50** to quickly move into the opening **115**.

(61) Because the follower **130** moves a double-stack of paintballs **50** into a single-stack at the contoured entry to the opening **115**, the paintballs **50** are more rapidly moved into the opening **115** than they would be if the follower **130** was pushing only a single stack of paintballs **50**.

(62) Before the magazine **100** is engaged to the adaptor **400**, the paintballs **50** are free to move around, spin, rotate, etc. within the wide interior **105** of the magazine **100**. When the magazine **100**

is engaged to the adaptor **400**, the spring-loaded follower **130** is released and urges against paintballs **50** to align the paintballs **50** in the double-stack, alternating pattern. The interior **105** and the magazine well **108** are shaped and configured to automatically align the paintballs **50** in the double-stack, alternating pattern upon release of the spring-loaded follower **130**. When the spring-loaded follower **130** is released, the spring-loaded follower **130** immediately places tension on the paintballs **50** and assembles the paintballs **50** in the double-stack, alternating pattern. In this aspect, when the spring-loaded follower **130** is released, the spring-loaded follower **130** travels toward the front of the magazine **100**.

(63) As such, it should be understood that the disclosure is not limited to the particular aspects described herein, but that various changes and modifications may be made without departing from the spirit and scope of this novel concept as defined by the following claims. Further, many other advantages of applicant's disclosure will be apparent to those skilled in the art from the above descriptions and the claims below.

Claims

1. A magazine and adapter for a paintball marker, comprising: a magazine configured to hold paintballs; an adapter configured to removably engage to a paintball marker; the magazine configured to engage to the adapter; the magazine comprised a spring-loaded follower, the spring-loaded follower having a locked configuration and an unlocked configuration, wherein in the locked configuration, the spring-loaded follower is fixed in a position relative to the magazine, and wherein in the unlocked configuration, the spring-loaded follower is configured to move relative to the magazine; wherein an engagement of the magazine in the locked configuration to the adapter transitions the spring-loaded follower to the unlocked configuration; wherein the magazine includes a magazine well, and the magazine well is contoured to hold paintballs in a double-stack pattern and to urge the paintballs into a single-stack pattern at an exit opening of the magazine; and wherein in the locked configuration, the paintballs are free to move or rotate in the magazine, and wherein in the unlocked configuration, the spring-loaded follower biases the paintballs in the single-stack pattern towards the exit opening of the magazine.
2. The magazine and adapter for a paintball marker according to claim 1, wherein in the locked configuration, the spring-loaded follower does not place tension on the paintballs in the magazine, and wherein in the unlocked configuration, the spring-loaded follower urges the paintballs in the magazine towards the exit opening of the magazine.
3. The magazine and adapter for a paintball marker according to claim 1, wherein a lower surface of the adapter engages an upper portion of the paintball marker in a removable engagement, wherein an upper surface of the adapter comprises a knob, and wherein the knob is configured to release the spring-loaded follower from the locked configuration.
4. The magazine and adapter for a paintball marker according to claim 3, wherein the knob is configured to fit into a channel of a lower wall of the magazine to engage a release switch for the spring-loaded follower to release the spring-loaded follower from the locked configuration.
5. The magazine and adapter for a paintball marker according to claim 4, wherein the engagement of the magazine in the locked configuration to the adapter engages the release switch to release the spring-loaded follower from the locked configuration.
6. The magazine and adapter for a paintball marker according to claim 5, wherein the magazine comprises a spring-loaded pin configured to hold the spring-loaded follower in the locked configuration, and the release switch is configured to retract the spring-loaded pin to unlock the spring-loaded follower from the locked configuration.
7. The magazine and adapter for a paintball marker according to claim 1, wherein the adapter comprises a catch mechanism, wherein the catch mechanism is configured to engage a catch opening in a lower wall of the magazine, wherein the adapter comprises a lever to disengage the

catch mechanism from the catch opening.

8. The magazine and adapter for a paintball marker according to claim 1, wherein the magazine includes a sliding door that is configured to retract relative to the magazine to cover and uncover an exit opening of the magazine.

9. The magazine and adapter for a paintball marker according to claim 8, wherein the engagement of the magazine to the adapter causes the sliding door to retract and uncover the exit opening to the magazine.

10. The magazine and adapter for a paintball marker according to claim 8, wherein an upper surface of the adapter includes a raised entry hole, and the raised entry hole includes walls configured to contact a front surface of the sliding door of the magazine.

11. The magazine and adapter for the paintball marker according to claim 1, wherein the magazine includes lateral extending members that extend outward from the magazine, and the lateral extending members are configured to slide under flanges of the adapter when the magazine is engaged to the adapter and moved toward a front of the adapter.

12. The magazine and adapter for a paintball marker according to claim 11, wherein the adapter comprises a catch mechanism, wherein the catch mechanism is configured to engage a catch opening in a lower wall of the magazine, wherein the catch mechanism is configured to prevent the magazine from sliding toward a rear of the adapter.

13. The magazine and adapter for a paintball marker according to claim 1, wherein the magazine comprises an upper wall, a middle wall, and a lower wall, and the upper wall and the middle wall define an interior to hold paintballs, and the lower wall comprises a door configured to cover and uncover an exit opening for the magazine.

14. The magazine and adapter for a paintball marker according to claim 13, wherein the lower wall houses a switch that actuates a spring-loaded pin to release a spring-loaded follower from the locked configuration.

15. The magazine and adapter for a paintball marker according to claim 1, wherein in the unlocked configuration, the magazine aligns the paintballs in a double-stack, alternating pattern.

16. The magazine and adapter for a paintball marker according to claim 15, wherein the magazine merges the double-stack, alternating pattern into a single-stack pattern as the paintballs approach the exit opening of the magazine.

17. The magazine and adapter for a paintball marker according to claim 16, wherein the paintballs pass through a feed tunnel of the adapter in a single-stack pattern.

18. The magazine and adapter for a paintball marker according to claim 1, wherein lateral walls of the magazine taper inwards towards a front of the magazine.

19. The magazine and adapter for a paintball marker according to claim 1, wherein the magazine well is horizontally contoured, and the magazine well is vertically contoured at the exit opening.

20. The magazine and adapter for a paintball marker according to claim 1, wherein the magazine well is horizontally contoured, and a front of the magazine well is vertically contoured.

21. The magazine and adapter for a paintball marker according to claim 1, wherein a lower surface of the adapter engages an upper portion of the paintball marker in a removable engagement.

22. The magazine and adapter for a paintball marker according to claim 1, wherein a lower surface of the adapter includes a groove configured to lockingly engage an upper portion of the paintball marker.

23. A magazine and adapter for a paintball marker, comprising: a magazine configured to hold paintballs; an adapter configured to removably engage to a paintball marker; the magazine configured to engage to the adapter; the magazine comprised a spring-loaded follower, the spring-loaded follower having a locked configuration and an unlocked configuration, wherein in the locked configuration, the spring-loaded follower is fixed in a position relative to the magazine, and wherein in the unlocked configuration, the spring-loaded follower is configured to move relative to the magazine; wherein an engagement of the magazine in the locked configuration to the adapter

transitions the spring-loaded follower to the unlocked configuration; wherein the magazine includes a magazine well, and the magazine well is contoured to hold paintballs in a double-stack pattern and to urge the paintballs into a single-stack pattern at an exit opening of the magazine; wherein in the locked configuration, the paintballs are free to move or rotate in the magazine, and wherein in the unlocked configuration, the spring-loaded follower biases the paintballs in the single-stack pattern towards the exit opening of the magazine, and wherein the magazine is horizontally contoured, wherein an interior width of the magazine is greater than a diameter of the paintballs and less than double the diameter of the paintballs.

24. A magazine and adapter for a paintball marker, comprising: a magazine configured to hold paintballs; an adapter configured to removably engage to a paintball marker; the magazine configured to engage to the adapter; the magazine comprised a spring-loaded follower, the spring-loaded follower having a locked configuration and an unlocked configuration, wherein in the locked configuration, the spring-loaded follower is fixed in a position relative to the magazine, and wherein in the unlocked configuration, the spring-loaded follower is configured to move relative to the magazine; wherein an engagement of the magazine in the locked configuration to the adapter transitions the spring-loaded follower to the unlocked configuration; wherein the magazine includes a magazine well, and the magazine well is contoured to hold paintballs in a double-stack pattern and to urge the paintballs into a single-stack pattern at an exit opening of the magazine; wherein in the locked configuration, the paintballs are free to move or rotate in the magazine, and wherein in the unlocked configuration, the spring-loaded follower biases the paintballs in the single-stack pattern towards the exit opening of the magazine, and wherein the spring-loaded follower travels parallel to a barrel or a firing direction of the paintball marker.

25. A magazine and adapter for a paintball marker, comprising: a magazine configured to hold paintballs; an adapter configured to removably engage to a paintball marker; the magazine configured to engage to the adapter; the magazine comprised a spring-loaded follower, the spring-loaded follower having a locked configuration and an unlocked configuration, wherein in the locked configuration, the spring-loaded follower is fixed in a position relative to the magazine, and wherein in the unlocked configuration, the spring-loaded follower is configured to move relative to the magazine; wherein an engagement of the magazine in the locked configuration to the adapter transitions the spring-loaded follower to the unlocked configuration; wherein the magazine includes a magazine well, and the magazine well is contoured to hold paintballs in a double-stack pattern and to urge the paintballs into a single-stack pattern at an exit opening of the magazine; wherein in the locked configuration, the paintballs are free to move or rotate in the magazine, and wherein in the unlocked configuration, the spring-loaded follower biases the paintballs in the single-stack pattern towards the exit opening of the magazine, and wherein the paintballs pass through the exit opening of the magazine and into an entry hole of the adapter, down a feed tunnel of the adapter, through an exit opening of the adapter, and to a loading opening of the paintball marker.
